

S. M. MEHRAB HOSSAIN JAYEED

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CAREER OBJECTIVE

Final-year Computer Science undergraduate seeking an AI Engineer role to build AI, Generative AI, and Agentic AI solutions for real-world use cases. Experienced in developing full-stack applications powered by LLM APIs and RAG pipelines, running end-to-end data pipelines (cleaning, transformation, feature engineering, statistical analysis), and applying interpretable machine learning to derive decision-support insights. Comfortable working with Python, Git/GitHub, and cloud-backed databases, with a structured, documentation-driven approach to prototyping, testing, and presenting AI solutions to non-technical stakeholders.

SKILLS

- **AI, GenAI & Agentic Systems:** LLM API integration (Groq, Gemini), RAG pipelines with vector search (Supabase/pgvector), Prompt Engineering, AI Chat/Behavioral Agents, Anthropic MCP
- **Data Analytics & ML:** Data Cleaning/Transformation, Exploratory Analysis, Feature Engineering, Interpretable/Explainable ML, K-means Clustering, SHAP, Model Evaluation, Scikit-learn
- **APIs & Automation:** REST APIs, webhooks (Supabase Realtime, FCM), CI/CD (GitHub Actions), Postman
- **Languages & Frameworks:** Python, JavaScript/TypeScript, Dart, C/C++, Next.js, React, Node.js, Flutter
- **Databases & Cloud:** Supabase (PostgreSQL), MongoDB, MySQL, Firebase, Docker, AWS Cloud Technical Essentials
- **Developer Tools:** Git, GitHub, VS Code, Vercel, Postman

PROJECTS

Double Gap Index (DGI) — Policy Intelligence Dashboard

[GitHub](#) | [Live Demo](#)

Next.js, TypeScript, Supabase, Python (ETL), MapLibre GL

- Built an interpretable policy dashboard scoring digital-access and physical-service-access gaps separately (never blended) across all **64 districts** of Bangladesh, flagging **37 districts (57.8%)** as compounded "double gap" cases using official BBS ICT Survey (2024–25) and OpenStreetMap data.
- Built a Python ETL pipeline computing normalized scores across **7 sub-indicators** per district, seeding a Supabase/PostgreSQL schema of **4 relational tables**, and applied K-means clustering with SHAP feature attribution to derive **4 policy archetypes**.
- Shipped an interactive Next.js frontend (choropleth map, 2x2 quadrant matrix, per-district dossiers) backed by a public REST API with **4 passing automated tests** validating scoring integrity and data-quality rules.

Focusnyx — Student Life OS & Cognitive Shield

[GitHub](#) | [Live Demo](#)

Next.js, Express.js, Supabase, Groq/Gemini API

- Built a full-stack student productivity ecosystem over **3–4 months**, combining a Next.js PWA, a Chrome extension (Manifest V3), and a Windows native companion app integrated via a shared backend API and real-time sync workflow.
- Implemented a two-way real-time focus-lock sync across all three clients, browser-level URL blocking, and low-level Win32 keyboard/window hooks for desktop lockdown.
- Integrated an AI Behavioral Coach and bilingual (English/Bangla) chatbot using Groq/Gemini APIs, an AI quiz generator from notes, and CGPA/SGPA momentum tracking with a Supabase/PostgreSQL backend, including a 3-tier productivity analytics dashboard.

UniShareSync — University Collaboration & Campus App

[GitHub](#) | [Live Demo](#)

Flutter, Supabase, FCM, Groq API

- Built a cross-platform Flutter app unifying academic resources, project collaboration, campus logistics, and peer commerce into a single role-based (Student/Faculty/Admin) ecosystem; secured **2nd place** at the Soft-

ware Project Showcase 2026.

- Integrated a Groq-powered AI Campus Assistant using RAG with vector-embedding search (Supabase/pgvector) over course documents for citation-backed, resource-grounded answering, plus a real-time collaborative Kanban board and whiteboard via Supabase Realtime webhooks.
- Implemented live GPS bus tracking with OpenStreetMap, a trust-scored CampusShare marketplace, and FCM push notification integration across the platform.

CERTIFICATIONS

- **CS50's Introduction to Artificial Intelligence with Python** — HarvardX, edX Verified Certificate
- **CS50x: Introduction to Computer Science** — HarvardX, edX Verified Certificate
- **Anthropic Model Context Protocol (MCP) Introduction** — Anthropic
- **AWS Cloud Technical Essentials** — Grameenphone Academy

HONORS & ACHIEVEMENTS

- **Datathon — ML Contest 2026:** Built an end-to-end predictive pipeline on a loan approval dataset including imputation, IQR outlier removal, feature encoding, scaling, and 20+ evaluated models tuned with RandomizedSearchCV.
- **2nd Place — Software Project Showcase 2026:** Secured second place presenting the UniShareSync platform.
- **Champion — Project Showcasing 2025:** Recognized for engineering and problem-solving excellence in a robotics/IoT showcase.

EDUCATION

Bachelor of Science in Computer Science & Engineering

Shanto-Mariam University of Creative Technology

Current Semester: 10th | CGPA: 3.93 / 4.00 (up to 9th semester)

Expected Graduation: 2026

Dhaka, Bangladesh

Higher Secondary Certificate (HSC)

Dhaka City College

Science Group | GPA: 5.00 / 5.00

2021

Dhaka, Bangladesh

Secondary School Certificate (SSC)

Armanitola Govt. High School

Science Group | GPA: 5.00 / 5.00

2019

Dhaka, Bangladesh

EXTRA-CURRICULAR ACTIVITIES

- **Volunteer**, SMUCT CSE Fest V3.0 (2026)
- **Class Representative (CR)**, CSE 31st Batch, SMUCT

SOFT SKILLS

Analytical Problem-Solving | Teamwork | Technical Documentation | Adaptability | Self-Learning

LANGUAGES

Bangla: Native | English: Upper-Intermediate (B2)